

Credit Analysis

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Recommendation

- Use the following variables to decide credit limit for both current and new customers:
 - Internal risk score
 - Salary
 - Additional credit for segment = 'Móveis' due to higher ticket
- Gather more data and rebuild the framework with
 - Split between models for new vs current customers.
 - Design a model specific for propensity to spend instead of using spend and segment.

Context

- Past 3.5 K contracts from 991 customers made between 2021-08 and 2024-08 were analyzed.
- 70% of the contracts for training vs 30% for testing.
- For simplification we only considered paid, only the contracts paid with no more than 60 days since due date, and used it as the target feature.

KPIs

0.70

Test AUC

40%

\$ Approval Rate

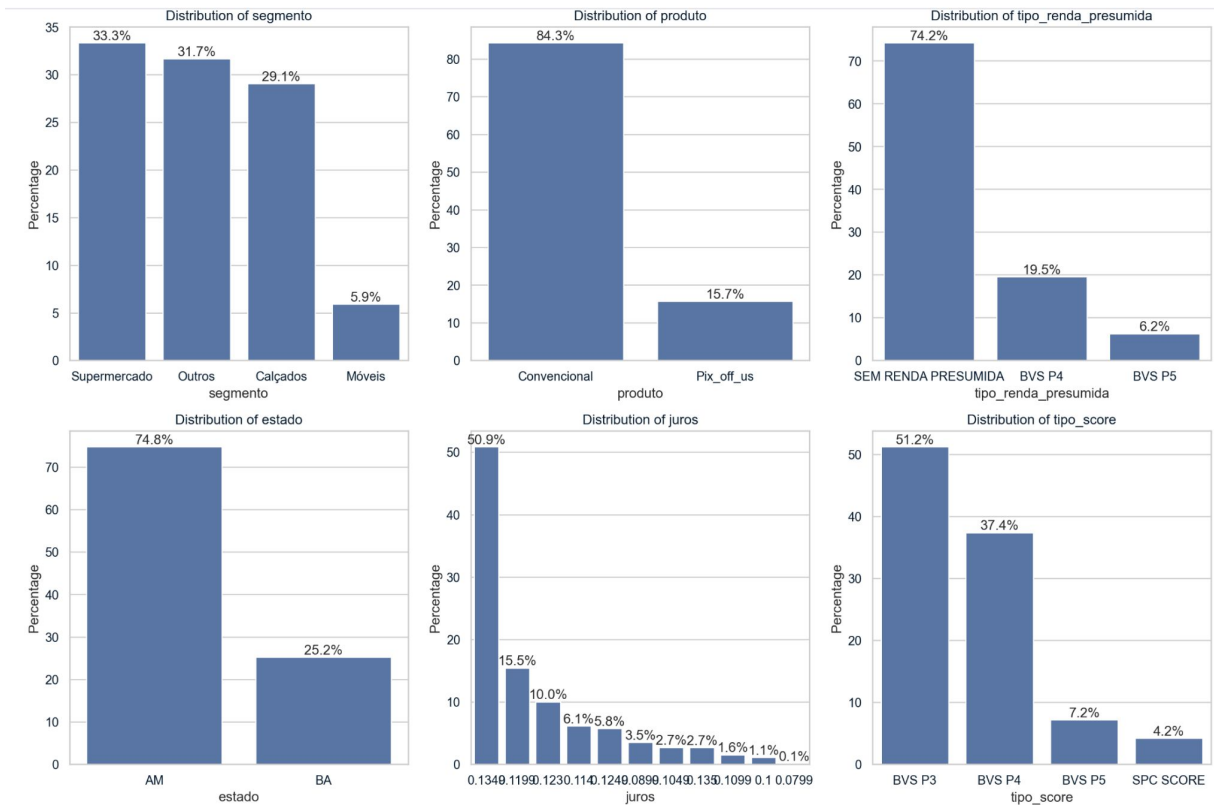
8.3%

Margin

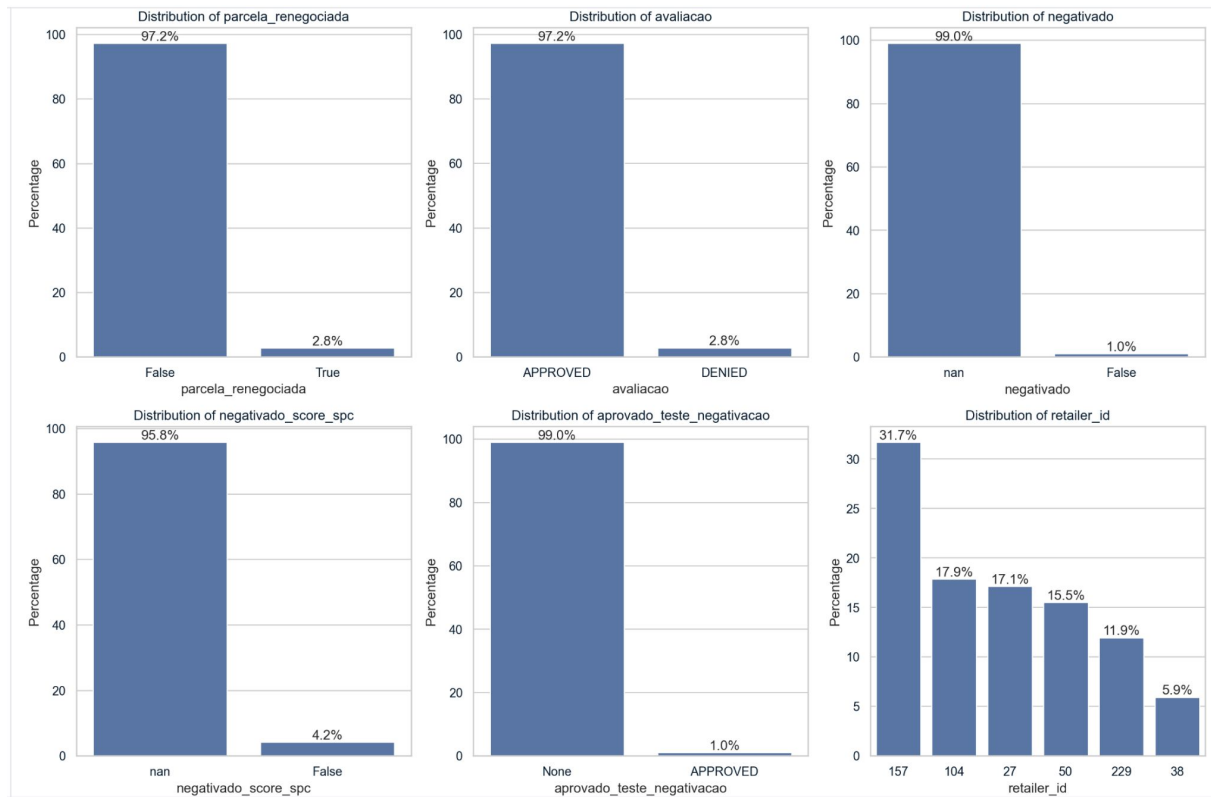
Data Format

1. **Client and Contract Information**
 - a. origination dates
 - b. installment details
 - c. client IDs,
 - d. contract maturity dates.
2. **Credit Performance Metrics:**
 - a. number of installments,
 - b. installment values,
 - c. interest rates (juros),
 - d. repayment behavior (e.g., paidValue, parcela_renegociada).
3. **Client Risk Profiles:**
 - a. credit scores (bvs_p3_score, bvs_p4_score, bvs_p5_score, spc_score),
 - b. presumed income (bvs_p3_renda_presumida, bvs_p4_renda_presumida, bvs_p5_renda_presumida)
 - c. indicadores (negativado and negativado_score_spc)
4. **Retailer and Product Details:**
 - a. Retailer (retailerId)
 - b. product involved in the transaction (produto).
 - c. Segments
 - d. States

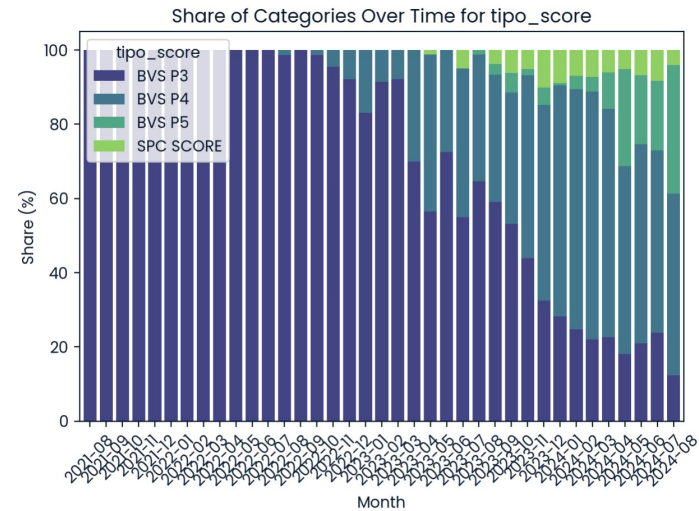
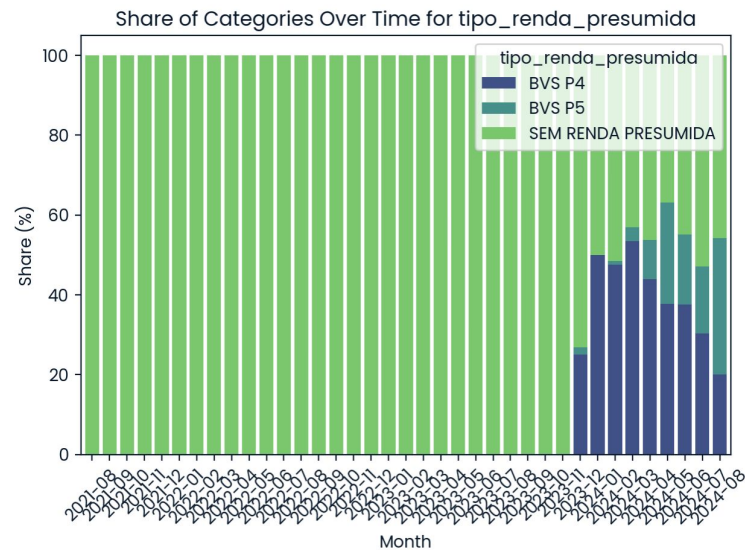
Few categories for most of segments



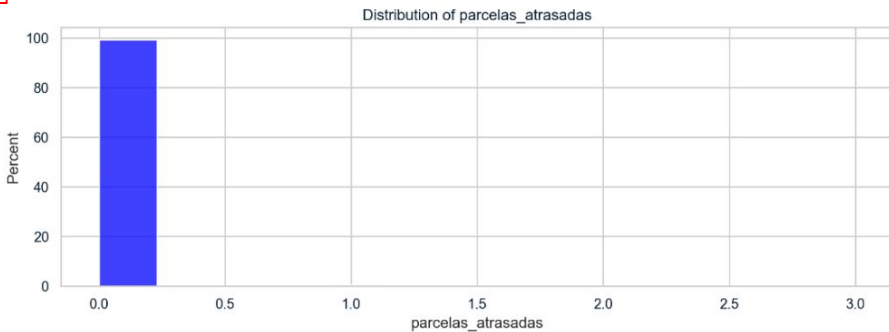
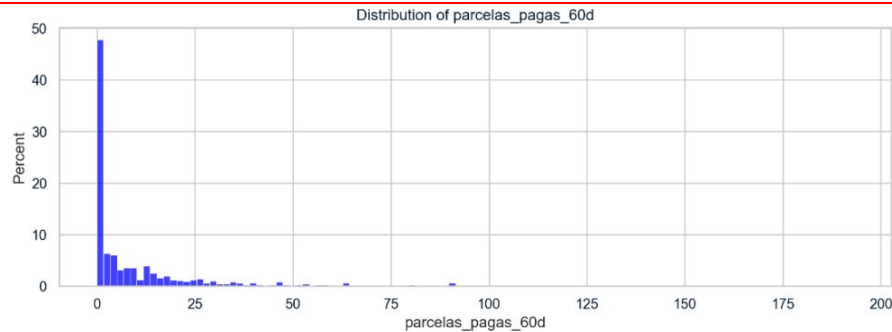
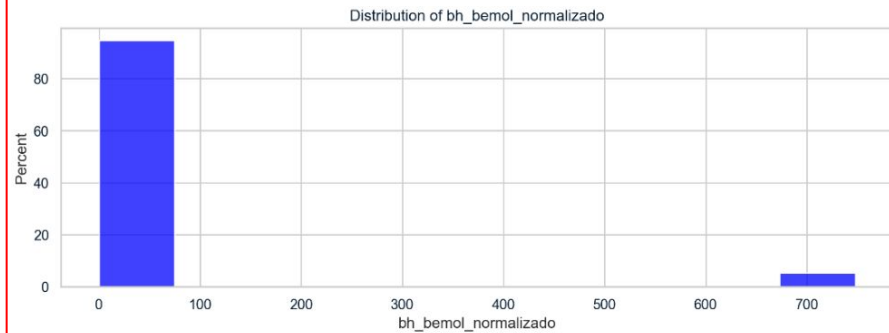
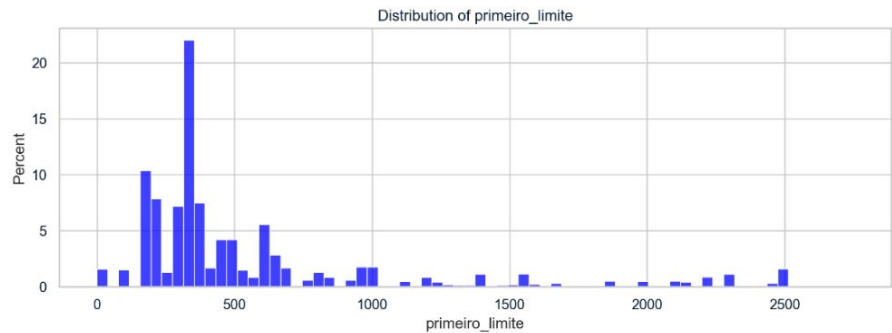
Skew distribution of flags



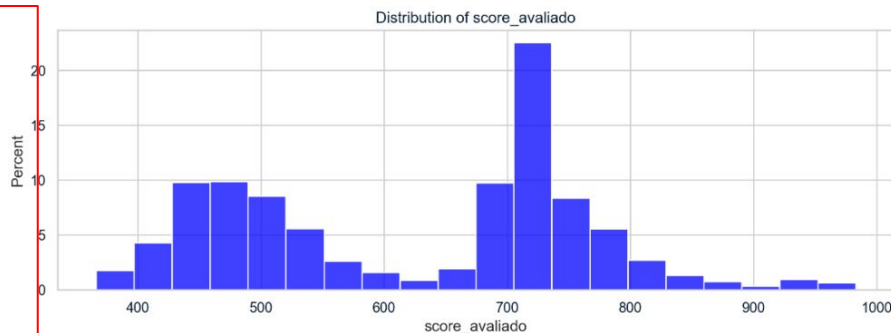
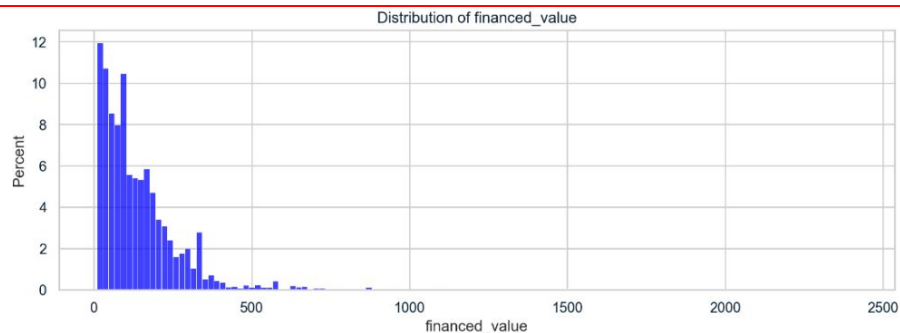
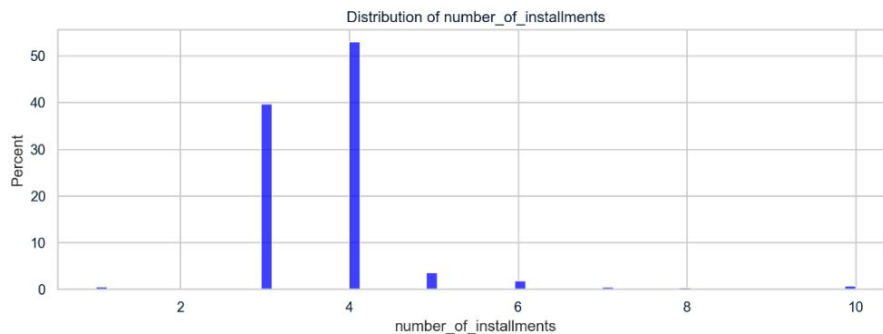
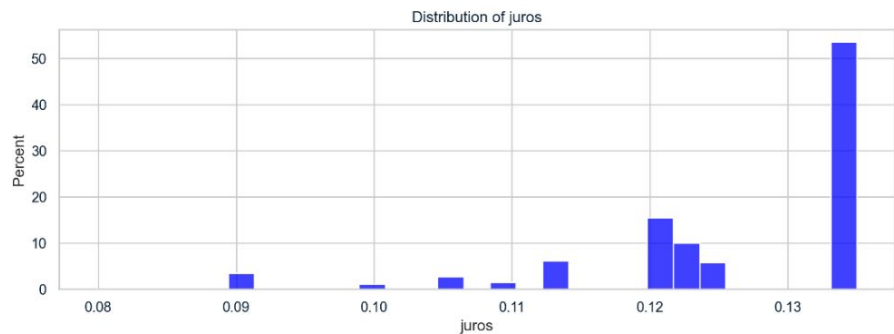
More data points incorporated recently, as newest scores and higher share with "renda presumida"



Most limits are lower than R\$ 500

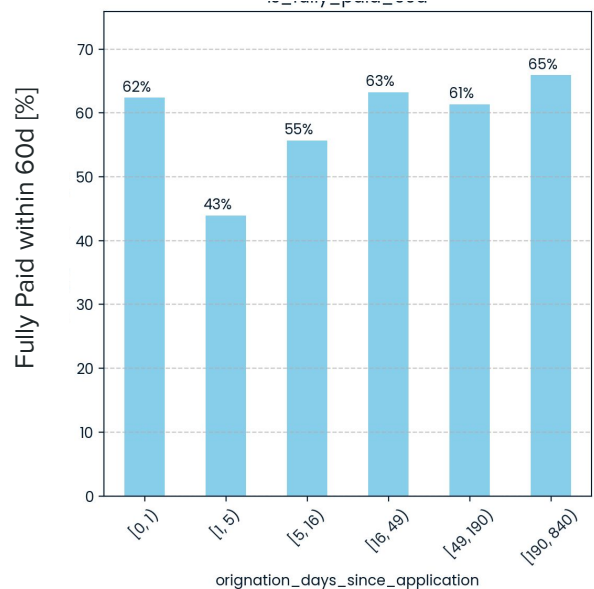


Low financed amounts, mostly lower than R\$ 500

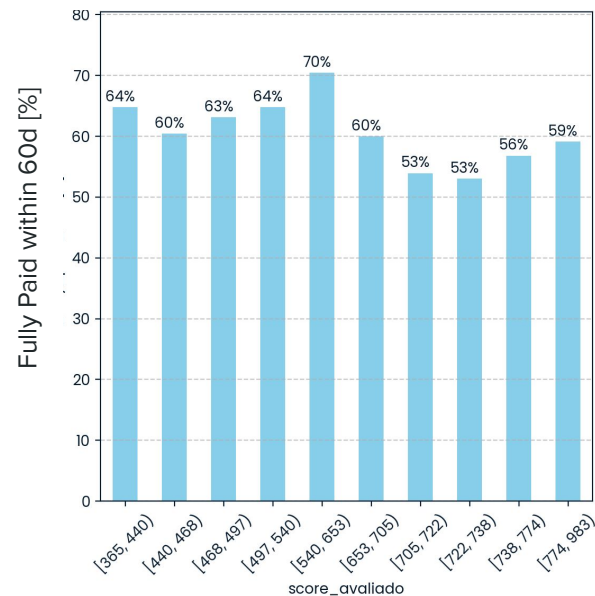


Relevant split in risk for origination days since application, but no relevant risk signal for credit score

Contracts Fully Paid¹ by Origination Days Since Application

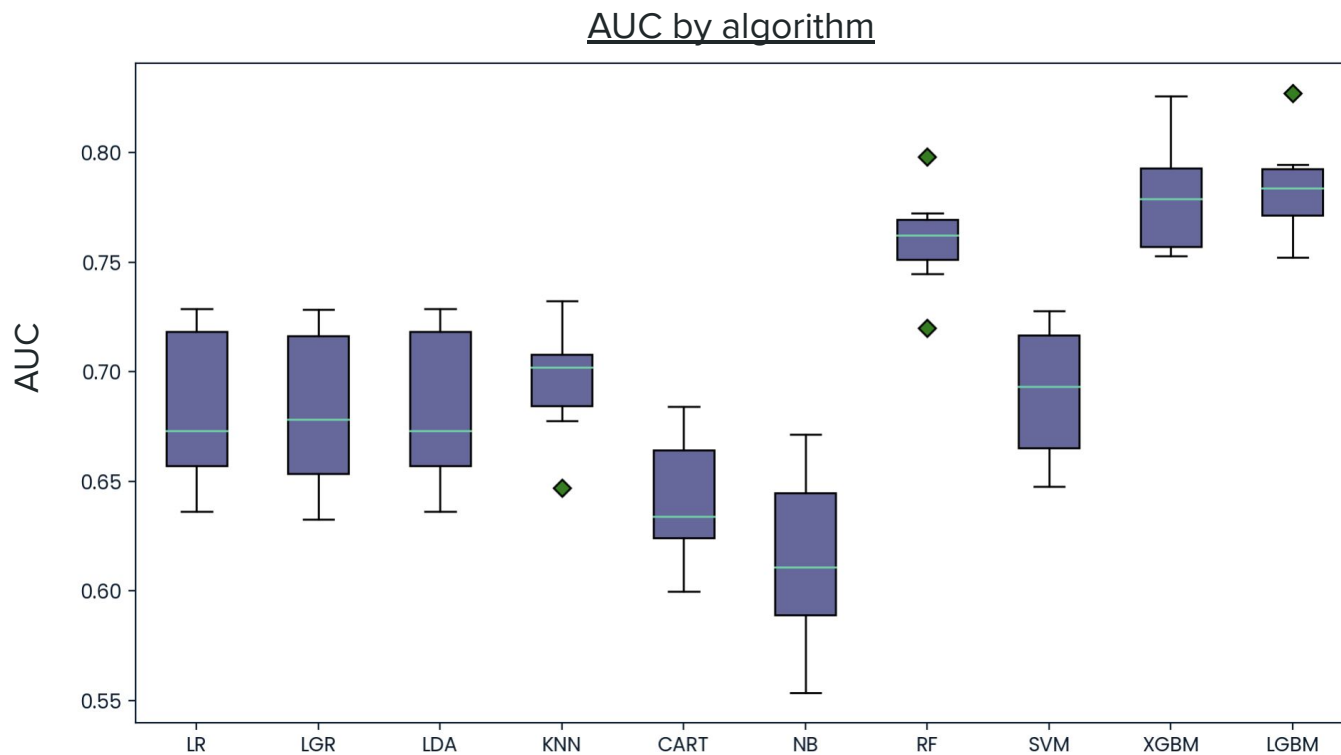


Contracts Fully Paid¹ by Credit Score



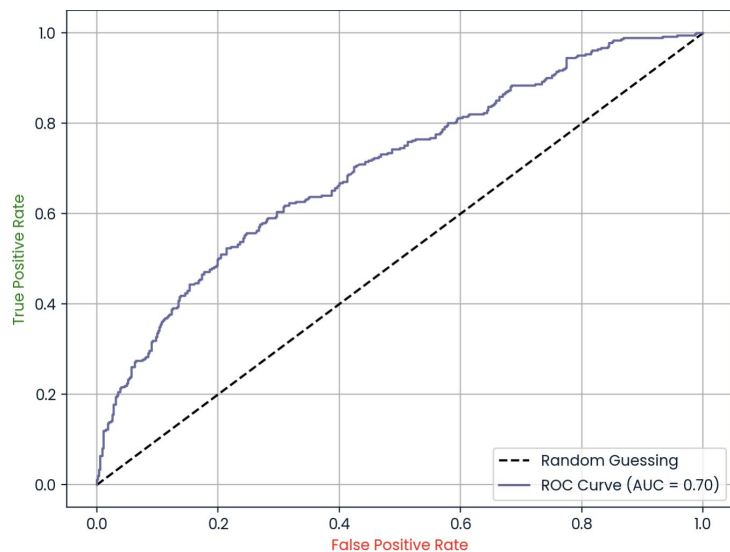
1. Contracts that had all their installments paid before 60 days after the due date

Best performance models were XGBM and LGBM by looking at their cross validation scores

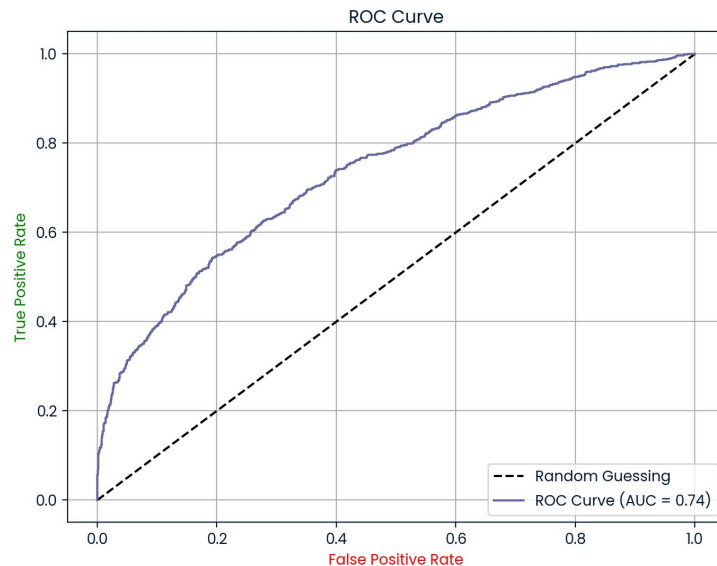


We achieved a similar score on test 0.7 AUC compared to training 0.75 AUC.

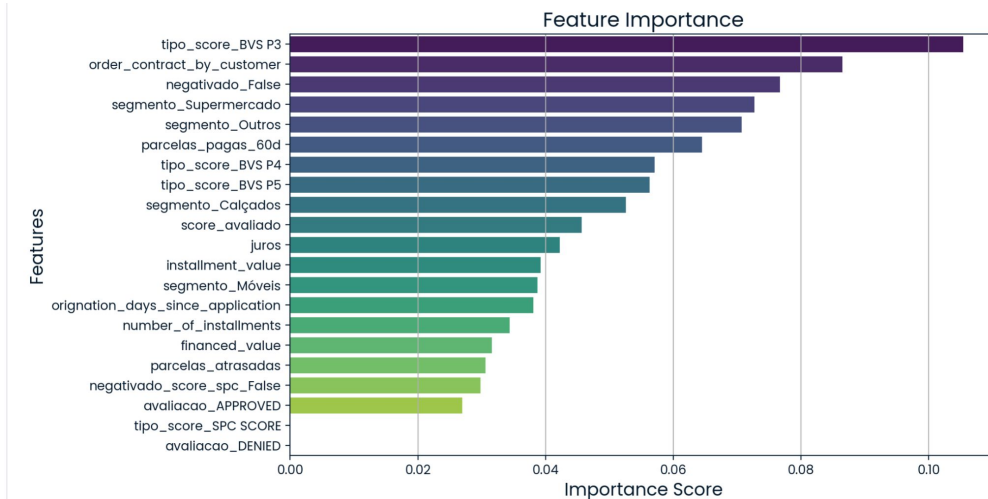
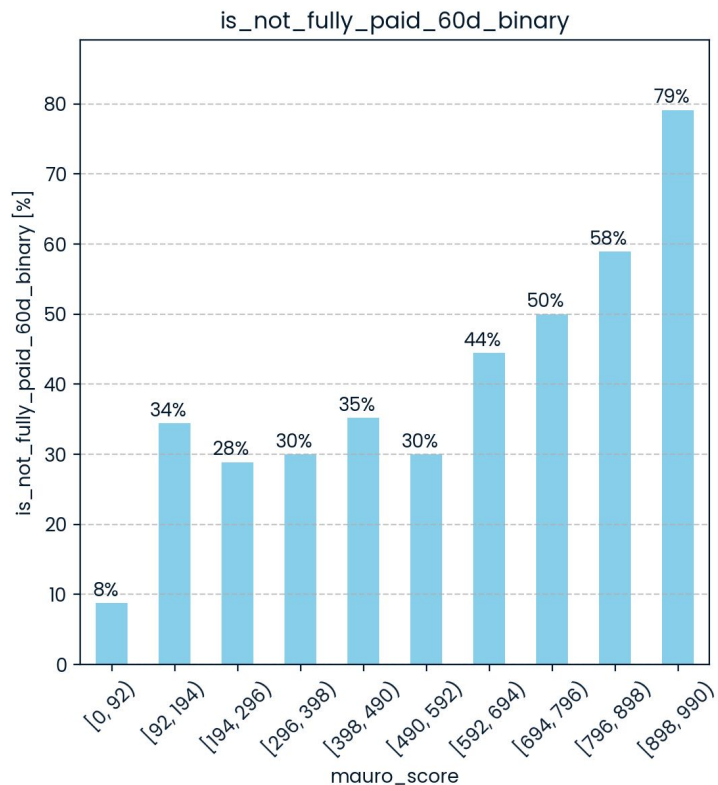
ROC Curve Test



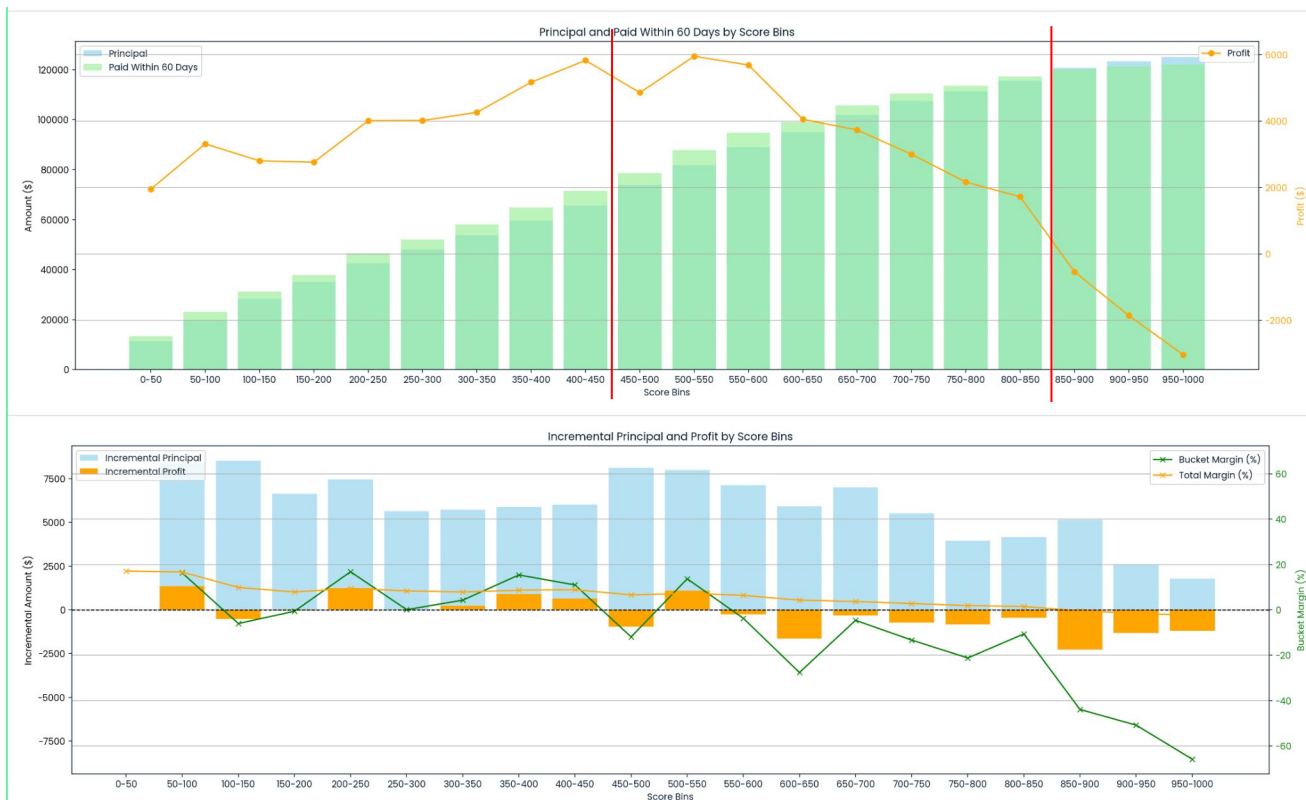
ROC Curve Training



Bottom 10% had 10x lower risk than top 10%, but still high risk for all buckets, also no feature had a high risk split as seen in the feature importance chart.



Considering the contracts performance, we would have achieved it's peak profitability for the 45% lower contracts, and break even with 85% of the contracts



The limit were chosen taking into account the 3 factors: Credit Score, Segment, and Salary

Limit =

Credit Score Factor



Segment



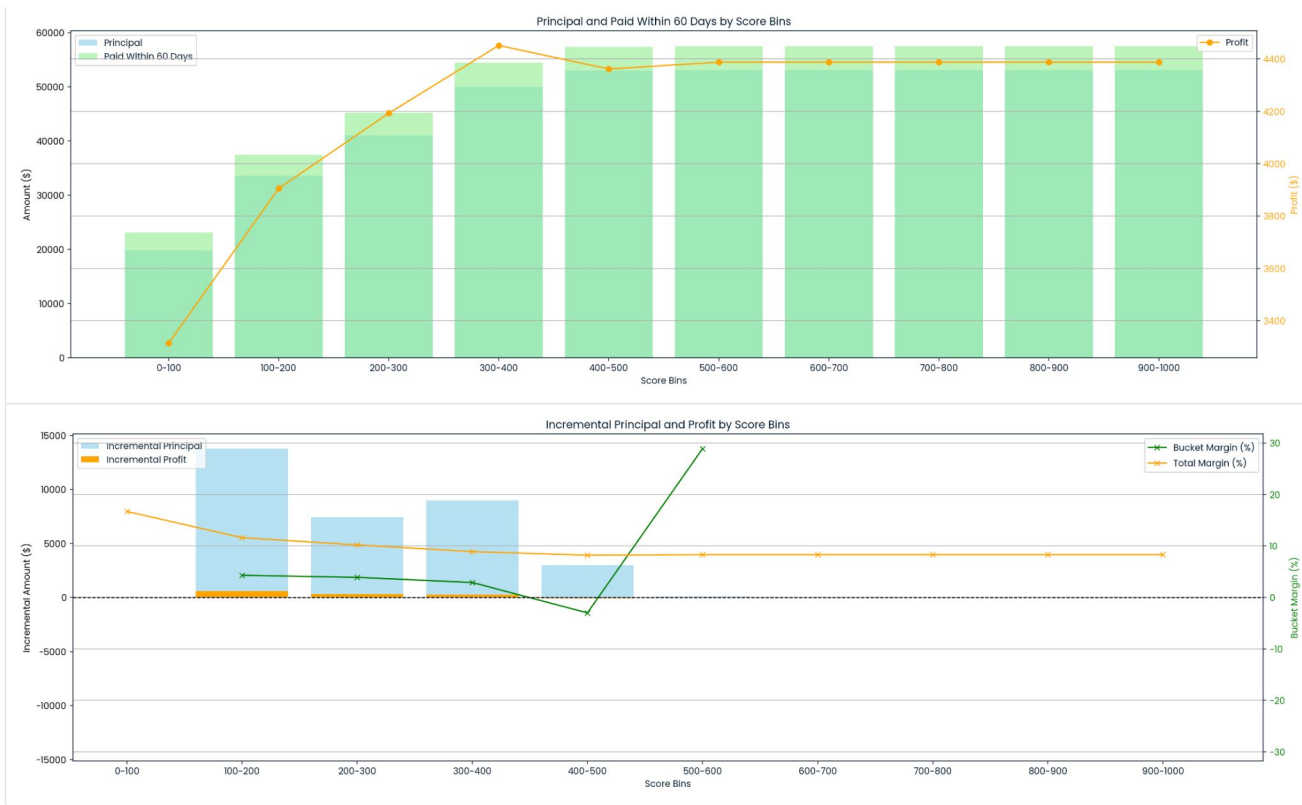
Salary Assumption

Max Credit Score	Factor
50	2
100	1
200	0.75
400	0.5
500	0.25
600	0.1
700	0.05
1000	0

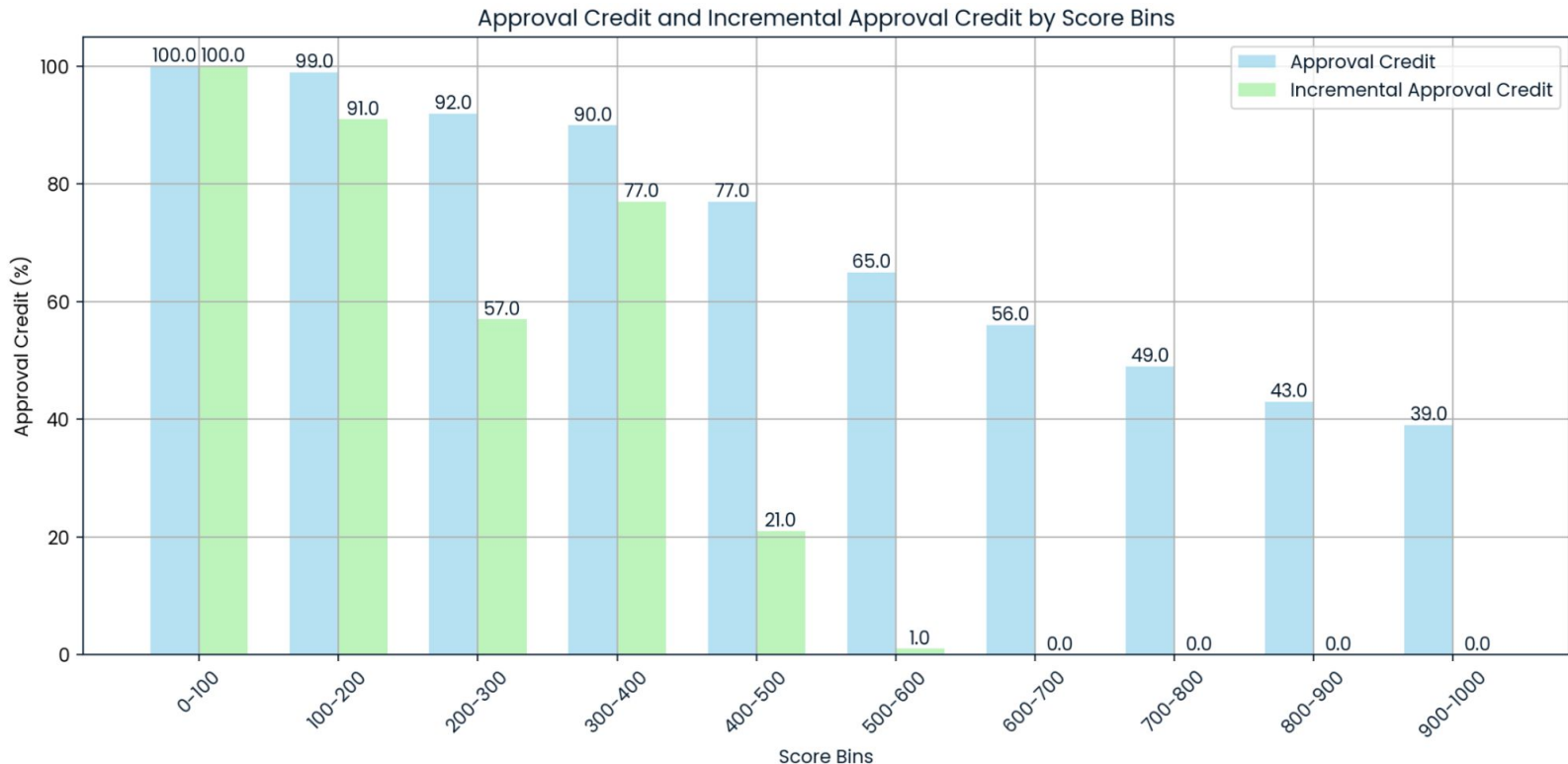
Segment	Factor
Móveis	1.5
All Others	1

Renda Presumida Avaliada	Value Considered 000's
Até R\$ 1.400	\$1,400
De R\$ 1.401 até R\$ 2.000	\$2,000
De R\$ 2.001 até R\$ 3.000	\$3,000
De R\$ 3.001 até R\$ 3.500	\$3,500
De R\$ 3.501 até R\$ 4.000	\$4,000
De R\$ 4.001 até R\$ 5.00	\$5,000
De R\$ 5.001 até R\$ 7.000	\$7,000
De R\$ 7.001 até R\$ 9.000	\$9,000
De R\$ 9.001 até R\$ 12.000	\$12,000
Acima de R\$ 12.000	\$15,000
Null	\$2,000

Considering only contracts with Principal + Inflight amount < Limit, we would have been profitable in \$4.500 in the test data



The approval rate for contracts by credit score, declines sharply and we decline pretty much all contracts from the top half riskier population



Thank you

AUC looking only at the first contract of each customer were too low to be used given the small sample size.

